

Abhishek Reddy Manam

Azure Data Engineer · Databricks · PySpark · ADLS Gen2 · ADF · FastAPI · REST API
Pipelines · Lakehouse · Delta Lake

+1 (484) 484-9961 | abhishekreddymanam@gmail.com | Wilmington, DE | [Abhishek Reddy Manam — LinkedIn](#)

PROFESSIONAL SUMMARY

Experienced Azure Data Engineer with 2+ years of proven experience designing and building scalable data pipelines, implementing data lake and lakehouse architectures, and delivering high-performance enterprise data integration solutions — with hands-on expertise in Azure Databricks, Python, PySpark, SQL, Azure Data Lake Storage Gen2, and Azure Data Factory. Develops automated ingestion frameworks for API-based and external data sources; builds REST API pipelines using Python and FastAPI for enterprise-scale data ingestion from structured and unstructured sources. Expert in implementing and maintaining data lakehouse architectures using Delta Lake with medallion (Bronze/Silver/Gold) design; optimizes and tunes pipeline performance for large-scale data processing with systematic Spark DAG analysis and ADF pipeline optimization. Strong understanding of relational and NoSQL databases; ensures data quality, reliability, and compliance with internal security and governance standards — with particular interest in cybersecurity and security data environments. Collaborates effectively with stakeholders across engineering, security, and architecture teams; supports agile delivery cycles and sprint-based development. Familiar with Azure Synapse Analytics and Microsoft Fabric as enterprise-scale lakehouse platforms. M.S. Information System Technology, Wilmington University (GPA 3.8, May 2026).

TECHNICAL SKILLS

Azure Databricks: Azure Databricks (required — hands-on): Clusters (job clusters, all-purpose, auto-scaling, instance pools), PySpark Notebooks & Workflows, Delta Lake (ACID, Z-ORDER, Liquid Clustering, MERGE INTO for CDC/SCD, schema evolution, time-travel), Databricks SQL Warehouses, Delta Live Tables (DLT), Unity Catalog (governance, access control), Cluster Policy Design, Cost Management, Databricks Asset Bundles CI/CD

Python & PySpark: Python (required — strong expertise): FastAPI (required — REST API development: endpoints, Pydantic validation, async handlers, OpenAPI docs, authentication middleware), Flask (familiar); PySpark (required — strong expertise): Distributed Data Processing, Spark DAG Analysis & CBO Optimization, Broadcast Joins, Partition Pruning, Predicate Pushdown, Shuffle Optimization, Structured Streaming, DataFrame & SQL APIs

SQL: SQL (required — strong expertise): Complex Queries, CTEs, Window Functions, Query Optimization, Execution Plan Analysis, Indexing & Partitioning, Stored Procedures; Databricks SQL, T-SQL, PostgreSQL, MySQL, Snowflake SQL; Relational & NoSQL Database Design (MongoDB, Cosmos DB, DynamoDB)

REST API Pipelines: REST API Ingestion Frameworks (required — proven experience): FastAPI-based API development, REST API connector design (OAuth 2.0, API key, Bearer token authentication), Pagination handling, Rate limiting & retry logic, JSON/XML payload parsing, Webhook event ingestion, External data source integration, API-to-ADLS Gen2 ingestion patterns, requests library, httpx

Azure Data Services: Azure Data Factory (ADF — required — hands-on): Automated Pipeline Design & Development, Linked Services (REST API, SFTP, databases), Copy Activity, Mapping Data Flows, Parameterized Pipelines, Triggers (scheduled, event-based, tumbling window), Integration Runtime (Azure & Self-Hosted), ADF Monitoring & CI/CD with Azure DevOps; Azure Data Lake Storage Gen2 (ADLS Gen2 — required): Hierarchical Namespace, Zone-Redundant Storage, Lifecycle Policies, Access Tiers, RBAC & ACL

Lakehouse Architecture: Data Lake & Lakehouse Architecture (required): Delta Lake Lakehouse Design, Medallion Architecture (Bronze/Silver/Gold), Enterprise-Scale Data Lakehouse Platform, Data Lake Organization & Governance, ADLS Gen2 Zone Design, Structured & Unstructured Data Ingestion, Azure Synapse Analytics (nice to have — familiar), Microsoft Fabric (nice to have — familiar): OneLake, Lakehouse, Fabric Pipelines

Performance Tuning: Pipeline Performance Optimization (required): Spark DAG Analysis, CBO Optimization, Z-ORDER Clustering & Liquid Clustering, Partition Pruning, Predicate Pushdown, Broadcast Joins, Shuffle Reduction, Delta Lake Compaction & Vacuum, ADF Pipeline Optimization, Databricks Cluster Right-Sizing, Azure Monitor Cost & Performance Dashboards

Data Quality & Security: Data Quality, Reliability & Compliance (required): Automated Validation Frameworks (dbt tests, Python validators), Schema Enforcement, Anomaly Detection, Data Governance Standards, RBAC & Access Controls (Azure AD, Unity Catalog), Managed Identity & Azure Key Vault, Encryption (at-rest & in-transit), HIPAA & GDPR Compliance, Security Data Environments (cybersecurity domain — nice to have: log ingestion, event data, SIEM feed patterns)

Databases: Relational Databases (required): SQL Server, PostgreSQL, MySQL, Azure SQL Database; NoSQL Databases (required): Azure Cosmos DB, MongoDB, DynamoDB, Redis; Relational vs. NoSQL trade-off understanding, Schema Design, Query Optimization

CI/CD & DevOps: Azure DevOps (required — agile delivery): CI/CD Pipelines (YAML), Repos (Git branching, PRs, code reviews), Sprint-Based Development, Agile/Scrum; GitHub Actions, Terraform (IaC — Azure Databricks workspace, ADLS Gen2, ADF provisioning), Docker, Databricks Asset Bundles; Azure Monitor & Log Analytics for observability

EDUCATION

M.S. Information System Technology · Wilmington University, Wilmington, DE · GPA 3.8/4.0 May 2026

B.S. Computer Science · Lovely Professional University, Punjab, India Jun 2023

PROFESSIONAL EXPERIENCE

Azure Data Engineer — Lead Contributor • DigiUniv Technologies Pvt. Ltd., India Jun 2023 – Jul 2024

- Designed and built automated data pipelines using Azure Databricks and Azure Data Factory — developed parameterized ADF pipelines with Integration Runtime, Copy Activity, and Mapping Data Flows; built Databricks Workflows with PySpark notebooks for distributed data processing; implemented and maintained data lake and lakehouse architectures using Delta Lake with medallion (Bronze/Silver/Gold) design in ADLS Gen2; ingested ~3TB of daily structured and unstructured data from 5 diverse source systems.
- Developed ingestion frameworks for API-based and external data sources — built REST API ingestion connectors using Python (requests, httpx) with OAuth 2.0 and Bearer token authentication, pagination handling, rate limiting, and retry logic; developed FastAPI-based data ingestion services with Pydantic request validation, async endpoints, and OpenAPI documentation; built webhook event ingestion pipelines; integrated enterprise security event feeds and external data sources into the lakehouse platform.
- Optimized and tuned pipeline performance for large-scale data processing — applied systematic Spark DAG analysis to identify CBO resource concentration; resolved data skew with salting and repartitioning; eliminated shuffles with broadcast joins and predicate pushdown; applied Z-ORDER clustering and Liquid Clustering reducing Databricks compute costs 30%; configured ADF pipeline parallelism and Integration Runtime optimization; implemented Delta Lake compaction and vacuum scheduling.
- Ensured data quality, reliability, and compliance with internal standards — implemented automated dbt schema tests and Python validation frameworks as mandatory CI/CD build gates; applied Azure RBAC, Managed Identity authentication, Unity Catalog governance, Azure Key Vault secrets management, HIPAA/GDPR compliance standards, and data lineage tracking; familiar with security data environments including log ingestion patterns and event-based data relevant to cybersecurity analytics use cases.
- Collaborated with stakeholders across engineering and security teams — partnered with data owners, security engineering, and architecture stakeholders to align pipeline design with enterprise standards; supported agile delivery cycles and sprint-based development using Azure DevOps; provisioned Azure infrastructure via Terraform IaC; maintained Git version control with structured code reviews; familiar with Azure Synapse Analytics and Microsoft Fabric as enterprise lakehouse platforms.

Data Engineer • DigiUniv Technologies Pvt. Ltd., India Jun 2022 – Jun 2023

- Built automated data pipelines using Azure Databricks and Azure Data Factory — developed REST API ingestion frameworks using Python with authentication, pagination, and retry logic; built FastAPI services for data ingestion endpoint development; implemented Delta Lake lakehouse architecture with medallion Bronze/Silver/Gold zones in ADLS Gen2; optimized PySpark pipeline performance with Spark DAG analysis and Z-ORDER clustering.
- Ensured data quality and collaborated cross-functionally — automated validation frameworks and schema enforcement; RBAC and Azure Key Vault security controls; supported agile sprint development with Azure DevOps; collaborated with security and engineering stakeholders; maintained Terraform IaC for Azure resource provisioning and GitHub Actions CI/CD for pipeline deployments.

KEY PROJECTS

Azure Lakehouse Platform — Databricks · ADLS Gen2 · ADF · REST API Ingestion · FastAPI | *Azure Databricks · Delta Lake · ADLS Gen2 · Azure Data Factory · Python · PySpark · FastAPI · REST API · Unity Catalog · Terraform · Azure DevOps · Azure Monitor*

- Designed and built a centralized enterprise data lakehouse — Azure Databricks with Delta Lake medallion (Bronze/Silver/Gold) architecture in ADLS Gen2 for structured and unstructured data; automated ADF pipelines for multi-source ingestion; REST API ingestion frameworks with Python (OAuth 2.0, pagination, rate limiting, retry logic); FastAPI-based data ingestion services with Pydantic validation and async endpoints; Unity Catalog governance with RBAC and column-level access controls.
- Optimized performance and ensured data quality — Spark DAG analysis and Z-ORDER clustering reducing compute costs 30%; ADF pipeline parallelism tuning; Azure Monitor and Log Analytics for observability; automated dbt and Python validation frameworks for data quality and compliance; Terraform IaC for all Azure resource provisioning; Azure DevOps CI/CD for sprint-based agile delivery; collaborated with engineering and security stakeholders throughout.

REST API Data Ingestion Framework — FastAPI · Python · Databricks · Security Data | *FastAPI · Python · REST APIs · OAuth 2.0 · Azure Databricks · ADLS Gen2 · Delta Lake · Azure Data Factory · Azure Key Vault · Managed Identity · GitHub Actions*

- Built enterprise REST API ingestion framework — FastAPI services with Pydantic request validation, OpenAPI documentation, async request handling, and authentication middleware; Python REST API connectors with OAuth 2.0 Bearer token authentication, pagination, rate limiting, dead-letter handling, and retry logic for external data sources including security event feeds and structured/unstructured enterprise data sources.
- Integrated API-ingested data into the Delta Lake lakehouse — structured landing zones in ADLS Gen2 Bronze layer; ADF pipeline triggers for automated ingestion scheduling; PySpark transformation in Databricks for normalization and Silver/Gold layer promotion; Azure Key Vault and Managed Identity for secure credential management; GitHub Actions CI/CD for automated deployment; demonstrated interest in cybersecurity data environments and security event data ingestion patterns.